Saekwang Nam

Assistant Professor in the Graduate School of Data Science
Kyungpook National University, 80 Daehak-ro, Daegu 41566, South Korea
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EDUCATION

2022 **Dr. rer. nat. in Computer Science**, Eberhard Karls University of Tübingen, Germany
The doctoral research has been conducted at Max Planck Institute for Intelligent Systems, Stuttgart, Germany
Advisor: Katherine J. Kuchenbecker, Ph.D.
Magna Cum Laude

2013 **M.S. in Computer Science**, University of California, San Diego, La Jolla, USA
Academic concentration: Artificial Intelligence

2011 **B.E. in Human & Mechanical Systems Engineering**, Kanazawa University, Kanazawa, Japan
Academic concentration: Mechatronics
Summa Cum Laude

EXPERIENCE

Kyungpook National University Mar. 2024 - present
Assistant Professor in the Graduate School of Data Science *Daegu, South Korea*

I am leading Robot Touch Data Laboratory, where I have focused on

- developing a multimodal tactile sensor for robot hands/grippers.
- building a data-driven touch classification model using a variety of ML models.

Bristol Robotics Laboratory & University of Bristol Nov. 2022 - Feb. 2024
Research Associate in Dexterous Robotics Group *Bristol, United Kingdom*

I am mainly involved in the Research Centre for Smart, Collaborative Industrial Robotics, funded by EPSRC, where I am working on

- a proprioceptive Fin-ray gripper based on the TacTip principle to estimate deformation, contact point, and depth.
- a ROS-based autonomous robot system featuring a UR-10 robot arm with a Robotiq gripper for object detection, pickup, and placement (Website).

Max Planck Institute for Intelligent Systems Sep. 2017 - Oct. 2022
Doctoral student *Stuttgart, Germany*

Title of the dissertation: Understanding the Influence of Moisture on Fingerpad-Surface Interactions

Theme I: Hardware developments for catching physical variables of human finger, for which I

- built an apparatus to measure three-dimensional finger contact forces, contact images, and fingerpad moisture (link, paper).
- improved the apparatus for better data synchronization, better resolution of fingerprint images, and better ambient temperature controls.

- developed a thin and transparent capacitive-type moisture sensor providing continuous moisture measurements based on laser lithography.

Theme II: Software skills such as image processing, simulations, and optimization

- I implemented a processing algorithm to extract the contact area from contact fingerprint images based on intensity gradient and adaptive threshold methods.
- I constructed a variable optimization system using particle swarm optimization to find the material properties of the human finger through repetitive finger-pressing simulations in COMSOL Multiphysics (paper).
- I have been implementing a fast parameter optimization algorithm to quickly extract a human finger's moisture level from optimized electrochemical impedance spectroscopy.

Electronics and Telecommunications Research Institute (ETRI) Apr. 2013 - Aug. 2017
Researcher *Daejeon, South Korea*

Theme I: Developing transparent and soft actuators, particularly focusing on

- Developing soft actuators based on electro-active polymers
- Figuring out the mechanical properties of dielectric elastomers
- Developing a bio-inspired varifocal lens and a thin-film mirror deformed by electrostatic force (paper).

Theme II: Connecting users to haptic applications that includes

- Developing a glass-plate actuator vibrating with the electrostatic force created between the stacked glass plates
- Building a graphic user interface to continuously present contact force measured from an optical waveguide-based transparent & flexible sensor array film (paper)
- Generating diverse input signals to haptic devices for the evaluation of a newly developed film-type sensor (paper).

Samsung Electronics Co., Ltd. Jul. - Sep. 2012
Software intern *Suwon, South Korea*

I was selected one of the worldwide interns as Samsung's global internship program, targeting excellent foreigners (including Koreans who study overseas) to work at the company in South Korea. In the Device Solutions department, I

- analyzed the traffic congestion of wafer-transferring vehicles in the semiconductor factories based on their log data files,
- computed the vehicle's maximum and average velocity in real-time and updated the information to the database.

DENSOTECHNO Co., Ltd. Aug. 2009
Intern *Obu, Japan*

- Learned a computer-aided design program (NX),
- Designed a radiator loaded on a Toyota's,
- Conducted simulations under diverse external disturbance conditions,
- Analyzed the areas of the radiator in which the disturbance could cause critical stresses.

AWARDS AND HONORS

2025 경북대 데이터사이언스대학원 'ICT 챌린지 2025' 에서 과학기술정보통신부 장관상 수상 (지도교수: 남세광; link)
Recipient of the Minister of Science and ICT Award at "ICT Challenge 2025" (Kyungpook National University, Supervised by Prof. Saekwang Nam, link)

2024 제2회 한국 햅틱스 학술대회의 미래 모빌리티 햅틱 기술 아이디어 공모전에서 인기상 수상 (지도교수: 남세광; link)
Popularity Award, Future Mobility Haptics Technology Idea Competition, the 2nd Korean Haptics Conference (Supervised by Prof. Saekwang Nam, link)

2024 Best Workshop Poster Award (with co-authors), ICRA 2024 ViTac Workshop

2021 Honorable Mention for the Best ToH Short Paper Award (with co-authors), World Haptics Conference

2021 Finalist for the Best Video Presentation Award (with co-authors), World Haptics Conference

2020 KIST Europe Scholarship Award (€1,000), Korean Scientists and Engineers Association in Germany

2020 Best Work-in-progress Poster Award (with co-authors), EuroHaptics Conference

2017 - 2022 A full funding for Ph.D. studies, International Max Planck Research School for Intelligent Systems

2015 Creativity and Innovation Award (with co-authors), ETRI

2015 Outstanding Paper Award (with co-authors), ETRI

2015 Outstanding Idea Award, ETRI

2014 Best Demonstration Award (with co-authors), Asia Haptics Conference

2014 The Most Popular Demonstration Award (with co-authors), Asia Haptics Conference

2014 Outstanding Paper Award (with co-authors), Institute of Control, Robotics and Systems domestic conference

2011 Hatakeyama Award, The Japan Society of Mechanical Engineers

2010 Kanazawa-kogyokai Award, Alumni Association in Engineering Departments of Kanazawa University

2009 Special Award, The Japan Society for Precision Engineering Hokuriku-Shinetsu Branch

2006 - 2011 A full funding for undergraduate studies, Korea-Japan Government Scholarship Program for the Students in Science and Engineering Departments (A stipend of ¥125,000/month and tuition fees)

PUBLICATIONS

Unpublished Papers (In Preparation and Under Review)

[U1] Masoud S. Bahraini, Ella-Mae Hubbard, Tiziana C. Callari, Isaiah Nassiuma, Paul Anandan, Pedro Ferreira, Cong Niu, Xiu Yan, Anas Shrinah, Kerstin Eder, Loong Yi Lee, **Saekwang Nam**, Jonathan Rossiter, Nathan Lepora, Seemal Asif, Iveta Eimontaite, Sarah Fletcher, Phil Webb, Angela Daly, Riccardo Vecellio Segate, Dariusz Ceglarek, Peter Kinnell, Niels Lohse: Human Behaviour and Intention Prediction for Industrial Human-Robot Collaboration: A Survey. *Submitted.*

[U2] **Saekwang Nam** and Katherine J. Kuchenbecker. The Importance of Sweat in Object Grasping: Why Is Dry Finger Slippery? *In preparation for submission.*

[U3] **Saekwang Nam**, Katherine J. Kuchenbecker, and the other collaborators. In vivo Transparent Sensor for the Continuous Hydration Measurement of Finger. *In preparation for submission.*

Journal Articles (Including Impact Factors in the year of publications)

[J1] **Saekwang Nam**, Bowen Deng, Loong Yi Lee, Jonathan M. Rossiter, and Nathan F. Lepora. Tacfinray: Soft tactile fin-ray finger with indirect tactile sensing for robust grasping. *IEEE Robotics and Automation Letters*, 11(3):2722–2729, 2026 *(Impact Factor = 5.3)*.

[J2] Taejun Kwon, Jieun Jang, and **Saekwang Nam**. Channel-attention 1d-cnns for real-time collision detection in human-robot interaction via audio spectral features. *International Journal of Control, Automation and Systems*, 23(11):3370–3382, 2025 *(Impact Factor = 2.9)*.

[J3] Loong Yi Lee, Silvia Terrile, **Saekwang Nam**, Tianhao Liang, Nathan Lepora, and Jonathan Rossiter. Fin-a-rays: Expanding soft gripper compliance via discrete arrays of flexible structures. *Soft Robotics*, 2025 *(Impact Factor = 6.1)*.

[J4] Chenghua Lu, Kailuan Tang, Xueming Hui, Haoran Li, **Saekwang Nam**, and Nathan F. Lepora. Stereotactip: Vision-based tactile sensing with biomimetic skin-marker arrangements. *IEEE/ASME Transactions on Mechatronics*, pages 1–13, 2025 *(Impact Factor = 7.3)*.

[J5] Paolo Susini, Giulia Pagnanelli, **Saekwang Nam**, Nathan F. Lepora, and Matteo Bianchi. Data-driven compliance discrimination via biomimetic soft optical tactile sensors: Implementation and benchmarking with a model-based approach. *IEEE Robotics and Automation Letters*, 10(8):8083–8090, 2025 *(Impact Factor = 5.3)*.

[J6] Taejun Kwon and **Saekwang Nam**. A preliminary study on softness optimization using machine learning for the digital twin of soft robots. *Journal of Institute of Control, Robotics and Systems*, 31(5):490–495, 2025 *(SCOPUS indexed)*.

[J7] Haoran Li, **Saekwang Nam**, Zhenyu Lu, Chenguang Yang, Efi Psomopoulou, and Nathan F. Lepora. Biotactip: A soft biomimetic optical tactile sensor for efficient 3d contact localization and 3d force estimation. *IEEE Robotics and Automation Letters*, 9(6):5314–5321, 2024 *(Impact Factor = 5.3)*.

[J8] Seung Koo Park, Sungryul Yun, Geonwoo Hwang, Meejeong Choi, Dong Wook Kim, Jong-Moo Lee, Bong Je Park, **Saekwang Nam**, Heeju Mun, Seongcheol Mun, Jeong Mook Lim, Eun Jin Shin, Ki-Uk Kyung, and Suntak Park. Highly contrastive, real-time modulation of light intensity by reversible stress-whitening of spontaneously formed nanocomposites: application to wearable strain sensors. *Journal of Materials Chemistry C*, 9(27):8496–8505, 2021 *(Impact Factor = 8.067)*.

[J9] **Saekwang Nam** and Katherine J. Kuchenbecker. Optimizing a viscoelastic finite element model to represent the dry, natural, and moist human finger pressing on glass. *IEEE Transactions on Haptics*, 14(2):303–309, 2021. Finalist for the Best Video Presentation Award, Second Honorable Mention for the Best ToH Short Paper Award at World Haptics Conference 2021 *(Impact Factor = 3.105)*.

[J10] **Saekwang Nam**, Yasemin Vardar, David Gueorguiev, and Katherine J. Kuchenbecker. Physical variables underlying tactile stickiness during fingerpad detachment. *Frontiers in Neuro-*

science, 14:235, 2020 *(Impact Factor = 4.677)*.

[J11] **Saekwang Nam**, Sungryul Yun, Jae Woong Yoon, Suntak Park, Seung Koo Park, Seongcheol Mun, Bongje Park, and Ki-Uk Kyung. A robust soft lens for tunable camera application using dielectric elastomer actuators. *Soft Robotics*, 5(6):777–782, 2018. PMID: 30156468 *(Impact Factor = 6.403)*.

[J12] Seongcheol Mun, Sungryul Yun, **Saekwang Nam**, Seung Koo Park, Suntak Park, Bong Je Park, Jeong Mook Lim, and Ki-Uk Kyung. Electro-active polymer based soft tactile interface for wearable devices. *IEEE Transactions on Haptics*, 11(1):15–21, 2018 *(Impact Factor = 2.757)*.

[J13] Suntak Park, Bongje Park, **Saekwang Nam**, Sungryul Yun, Seung Koo Park, Seongcheol Mun, Jeong Mook Lim, Yeonghwa Ryu, Seok Ho Song, and Ki-Uk Kyung. Electrically tunable binary phase fresnel lens based on a dielectric elastomer actuator. *Optics Express*, 25(20):23801–23808, 2017 *(Impact Factor = 3.356)*.

[J14] Seung Koo Park, Young-Je Kwark, **Saekwang Nam**, Jaehyun Moon, Dong Wook Kim, Suntak Park, Bongje Park, Sungryul Yun, Jeong-Ik Lee, Byounggon Yu, et al. A variation in wrinkle structures of uv-cured films with chemical structures of prepolymers. *Materials Letters*, 199:105–109, 2017 *(Impact Factor = 2.687)*.

[J15] Sungryul Yun, Suntak Park, **Saekwang Nam**, Bongje Park, Seung Koo Park, Seongcheol Mun, Jeong Mook Lim, and Ki-Uk Kyung. An electro-active polymer based lens module for dynamically varying focal system. *Applied Physics Letters*, 109(14):141908, 2016 *(Impact Factor = 3.411)*.

[J16] Seung Koo Park, Young-Je Kwark, **Saekwang Nam**, Suntak Park, Bongje Park, Sungryul Yun, Jaehyun Moon, Jeong-Ik Lee, Byounggon Yu, and Ki-Uk Kyung. Wrinkle structures formed by formulating uv-crosslinkable liquid prepolymers. *Polymer*, 99:447–452, 2016 *(Impact Factor = 3.684)*.

[J17] **Saekwang Nam**, Suntak Park, Sungryul Yun, Bongje Park, Seung Koo Park, and Ki-Uk Kyung. Structure modulated electrostatic deformable mirror for focus and geometry control. *Optics Express*, 24(1):55–66, 2016 *(Impact Factor = 3.307)*.

[J18] Bong Je Park, Suntak Park, Sungryul Yun, **Saekwang Nam**, and Ki-Uk Kyung. A transparent visuo-haptic input device with optical waveguide based thin film display, sensor and surface actuator. *Sensors and Actuators A: Physical*, 233:47–53, 2015 *(Impact Factor = 2.201)*.

[J19] Sungryul Yun, Suntak Park, Bongje Park, **Saekwang Nam**, Seung Koo Park, and Ki-Uk Kyung. A thin film active-lens with translational control for dynamically programmable optical zoom. *Applied Physics Letters*, 107(8):081907, 2015 *(Impact Factor = 3.142)*.

[J20] Suntak Park, Bong Je Park, Sungryul Yun, **Saekwang Nam**, Seung Koo Park, and Ki-Uk Kyung. Thin film display based on polymer waveguides. *Optics express*, 22(19):23433–23438, 2014 *(Impact Factor = 3.488)*.

[J21] Sungryul Yun, Suntak Park, Bongje Park, Youngsung Kim, Seung Koo Park, **Saekwang Nam**, and Ki-Uk Kyung. Polymer-waveguide-based flexible tactile sensor array for dynamic response. *Advanced Materials*, 26(26):4474–4480, 2014 *(Impact Factor = 17.493)*.

[J22] Ki-Uk Kyung, Sungryul Yun, Suntak Park, Bongje Park, Seung Koo Park, **Saekwang Nam**, Harsha Prahlad, and Philip von Guggenberg. Flexible visuo-haptic display. *Journal of Korea*

Robotics Society, 8:156–163, 2013 *(No Impact Factor Available).*

Peer-Reviewed Conference Papers, posters, and demo presentations

[C1] Junhui Lee, Hyosung Kim, and **Saekwang Nam**. Tactiverse: Generalizing multi-point tactile sensing in soft robotics using single-point data. In *2026 IEEE 8th International Conference on Soft Robotics (RoboSoft)*, 2026 *(Accepted)*

[C2] Hyosung Kim, Junhui Lee, and **Saekwang Nam**. Real-time multimodal tactile sensor with visual and auditory feedback. In *2025 IEEE-RAS 24th International Conference on Humanoid Robots, Dexterous Humanoid Manipulation Workshop*, 2025

[C3] Loong Yi Lee, Silvia Terrile, Ajmal Roshan, Tianqi Yue, **Saekwang Nam**, and Jonathan Rossiter. Down the rabbit hole: Exploiting airflow interactions for morphologically intelligent retracting vacuum grippers. In *2024 IEEE 7th International Conference on Soft Robotics (RoboSoft)*, pages 870–875, 2024

[C4] **Saekwang Nam***, Toby Jack*, Loong Yi Lee, and Nathan F. Lepora. Softness prediction with a soft biomimetic optical tactile sensor. In *2024 IEEE 7th International Conference on Soft Robotics (RoboSoft)*, pages 121–126, 2024

[C5] **Saekwang Nam**, Loong Yi Lee, Max Yang, Naoki Shitanda, Jonathan Rossiter, and Nathan Lepora. Vision and tactile pose identification for picking a target without collision. Poster presented at "ViTac" workshop held at ICRA 2023 Conference, May 2023

[C6] **Saekwang Nam**, David Gueorguiev, and Katherine J. Kuchenbecker. Finger contact during pressing and sliding on a glass plate. Poster presented at "Skin mechanics and its role in manipulation and perception" workshop held at EuroHaptics, May 2022

[C7] **Saekwang Nam** and Katherine J. Kuchenbecker. Sweat softens the outermost layer of the human finger pad: evidence from simulations and experiments. Work-in-progress poster presented at EuroHaptics Conference, September 2020. **Award for the Best Work-in-progress poster**

[C8] **Saekwang Nam** and Katherine J. Kuchenbecker. Understanding the pull-off force of the human fingerpad. Work-in-progress paper (2 pages) presented at the IEEE World Haptics Conference (WHC), July 2019

[C9] Maria-Paola Forte, Rachael L'Orsa, Mayumi Mohan, **Saekwang Nam**, and Katherine J. Kuchenbecker. The haptician and the alphamonsters. Student Innovation Challenge on Implementing Haptics in Virtual Reality Environment presented at the IEEE World Haptics Conference, July 2019. Maria Paola Forte, Rachael L'Orsa, Mayumi Mohan, and Saekwang Nam contributed equally to this publication

[C10] Seongcheol Mun, Sungryul Yun, **Saekwang Nam**, Seung-Koo Park, and Ki-Uk Kyung. Initial progress toward a surface morphable tactile interface. In Shoichi Hasegawa, Masashi Konyo, Ki-Uk Kyung, Takuya Nojima, and Hiroyuki Kajimoto, editors, *Haptic Interaction*, pages 113–114, Singapore, 2018. Springer Singapore

[C11] **Saekwang Nam**, Suntak Park, Sungryul Yun, Bong Je Park, Seung Koo Park, and Ki-Uk Kyung. Influence on deformed shape of electrostatic deformable mirror by the geometry of electrode. In *Conference on Lasers and Electro-Optics*, page JTh2A.7. Optical Society of America, 2016

[C12] Suntak Park, Jin Tae Kim, **Saekwang Nam**, Bong Je Park, Sungryul Yun, Seung Koo Park, and Ki-Uk Kyung. Pressure sensor using vertical coupling of optical waveguides. In *Conference on Lasers and Electro-Optics*, page JTu5A.144. Optical Society of America, 2016

[C13] **Saekwang Nam**, Suntak Park, Sungryul Yun, Bong Je Park, Seung Koo Park, Mijeong Choi, and Ki-Uk Kyung. *Highly Flexible and Transparent Skin-like Tactile Sensor*, pages 187–189. Springer Japan, Tokyo, 2015

[C14] **Saekwang Nam** and Tetsuyou Watanabe. The development of robot hand capable of expressing the movement of the palm(biorobotics). *Proceedings of the robotics and mechatronics conference in Japan*, 2011:1P1–P04(1–4), May 2011

PATENTS

[P1] Seung Koo Park, Sung Ryul Yun, Ki Uk Kyung, Mi Jeong Choi, Bong Je Park, Suntak Park, **Saekwang Nam**, Seongcheol Mun, Eun Jin Shin, Jeong Mook Lim, et al. Light transmittance control film and composition for the light transmittance control film, March 30 2023. US Patent 2023/0,103,068

[P2] Ki-uk Kyung, Bong Je Park, Sang-youn Kim, **Saekwang Nam**, Sun Tak Park, and Sung Ryul Yun. Auto focusing device, May 12 2020. US Patent 10,649,116

[P3] Ki-uk Kyung, **Saekwang Nam**, Bong Je Park, Sun Tak Park, and Sung Ryul Yun. Imaging apparatus with adjustable lens and method for operating the same, December 3 2019. US Patent 10,495,843

[P4] Ki Uk Kyung, **Saekwang Nam**, Sung Ryul Yun, Bong Je Park, Sun Tak Park, and Seung Koo Park. Varifocal lens module, July 2 2019. US Patent 10,338,280

[P5] Ki-uk Kyung, Bong Je Park, Sang-youn Kim, **Saekwang Nam**, Sun Tak Park, and Sung Ryul Yun. Auto focusing device, June 26 2018. US Patent 10,007,034

[P6] Sun Tak Park, Ki Uk Kyung, Sae Kwang Nam, Bong Je Park, Seung Koo Park, and Sung Ryul Yun. Optical imaging device, June 26 2018. US Patent 10,009,546

[P7] Sun Tak Park, Ki-uk Kyung, **Saekwang Nam**, Bong Je Park, Seung Koo Park, and Sung Ryul Yun. Image processing apparatus and control method thereof, June 5 2018. US Patent 9,992,433

[P8] Sung-Ryul Yun, Ki-uk Kyung, Sun-tak Park, Bong-Je Park, and **Saekwang Nam**. Microlens array and method for fabricating thereof, February 6 2018. US Patent 9,885,874

[P9] Sun Tak Park, Ki Uk Kyung, Sung Ryul Yun, **Saekwang Nam**, and Bong Je Park. Display apparatus and manufacturing method thereof, December 12 2017. US Patent 9,841,621

[P10] Ki-uk Kyung, Sung-Ryul Yun, Sun-tak Park, Bong-Je Park, and **Saekwang Nam**. Shape-variable optical element, August 29 2017. US Patent 9,746,587

[P11] Sung Ryul Yun, Ki Uk Kyung, **Saekwang Nam**, Bong Je Park, and Sun Tak Park. Artificial muscle, May 30 2017. US Patent 9,662,197

[P12] Sun-tak Park, Bong-Je Park, Ki-uk Kyung, Sung-Ryul Yun, and **Saekwang Nam**. Active diffuser for reducing speckle and laser display device having active diffuser, May 2 2017. US Patent 9,638,928

[P13] Sung Ryul Yun, Ki Uk Kyung, Sun Tak Park, **Saekwang Nam**, and Bong Je Park. Thin active optical zoom lens and apparatus using the same, April 4 2017. US Patent 9,612,362

[P14] **Saekwang Nam**, Sun Tak Park, Ki Uk Kyung, Bong Je Park, and Sung Ryul Yun. Active reflective lens and apparatus using the same, September 15 2016. US Patent App. 15/065,856

[P15] Sun Tak Park, Ki Uk Kyung, Bong Je Park, Sung Ryul Yun, and **Saekwang Nam**. Variable fresnel lens, April 12 2016. US Patent 9,310,532

[P16] **Saekwang Nam**, Ki-uk Kyung, Sung-Ryul Yun, Sun-tak Park, and Bong-Je Park. Shape-variable optical element and optical read/write device including the same, July 28 2015. US Patent 9,091,810

[P17] Sun Tak Park, Sung Ryul Yun, Bong Je Park, **Saekwang Nam**, and Ki Uk Kyung. Reflective varifocal lens and imaging system including the same, August 20 2015. US Patent App. 14/618,282

[P18] **Saekwang Nam**, Ki Uk Kyung, Sung Ryul Yun, Bong Je Park, and Sun Tak Park. Tunable lens system, July 23 2015. US Patent App. 14/600,369

[P19] Ki-uk Kyung, Sung-Ryul Yun, Sun-tak Park, Bong-Je Park, and **Saekwang Nam**. Variable-shape optical element, February 12 2015. US Patent App. 14/316,083

MEDIA HIGHLIGHTS

October 2025	Selected and featured as a 'Young Robot Engineer' by Robot News for leading innovations in robotic tactile data and haptics (Article link).
August 2025	Donga Science featured his expert commentary on the critical role of multi-modal data in tactile sensing research (News link).
June 2014	The innovative polymer waveguide-based tactile sensor array received national media coverage on KBS (News link).

GRANTS AND CONTRACTS

Current

1. 멀티모달 비전기반 촉각센싱에 의한 로봇의 물체 조작 기술 향상 – 2710019310
   과기정통부 - 개인기초연구(과기정통부)(R&D) - 글로벌 매칭형(영국)
   Role: 연구책임(PI) — PI: Saekwang Nam (KNU)
   Management Agency: 한국연구재단 (NRF) — Funding to Nam's Lab: ∼360M KRW
   Dates: Oct 1, 2024 – Sep 30, 2027

2. 데이터 기반 지능형 모빌리티 연구센터 – 2710035850
   과기정통부 - 정보통신방송혁신인재양성 (대학ICT연구센터)
   Role: 참여연구원(Co-investigator) — PI: Nakhoon Baek (KNU)
   Management Agency: 정보통신기획평가원 (IITP) — Funding to Nam's Lab: ∼450M KRW
   Dates: Jul 1, 2024 – Dec 31, 2031

3. GLOW-AI 혁신인재양성 교육연구단
   과기정통부 - 4단계 두뇌한국(BK)21사업 (혁신인재 양성사업)
   Role: 참여연구원(Co-investigator) — PI: Joonho Kwon (Pusan National University)
   Management Agency: 한국연구재단 (NRF) — Funding to Nam's Lab: ∼300M KRW
   Dates: Mar 1, 2025 – Feb 28, 2026 — Note: 3책5공 제외사업

4. 지역산업 혁신을 위한 지역 수요 중심 데이터사이언스 융합인재 양성사업 – 2710065097
   과기정통부 - 과학기술혁신인재양성(R&D) (데이터사이언스융합인재양성)
   Role: 참여연구원(Co-investigator) PI: Taeok Jung (KNU)
   Management Agency: 한국연구재단 (NRF) Funding to Nam's Lab: ∼300M KRW
   Dates: Apr 1, 2023 – Dec 31, 2029 Note: 3책5공 제외사업

Completed

1. 차세대 다중감각 통합 소셜 로봇 제어를 위한 범용 인공지능 플랫폼 개발 – 2710085246
   과기정통부 - 인간지향적 차세대 도전형 AI기술 개발 (핵심전력R&D)
   Role: 공동연구책임(Co-PI) PI: Yong-Goo Shin (Korea University)
   Management Agency: 정보통신기획평가원 (IITP) Funding to Nam's Lab: 30M KRW
   Dates: Apr 1, 2025 – Dec 31, 2025
2. 경북대학교 글로벌 매칭 Lab 지원사업
   경북대학교 국제처 - 국제협력센터
   Role: 연구책임(PI) PI: Saekwang Nam (KNU)
   Management Agency: 경북대학교 (KNU) Funding to Nam's Lab: ∼5M KRW/
   Dates: Oct 1, 2025 – Jan 31, 2026 Note: 3책5공 제외사업
3. 손가락의 특성에 기반한 로봇 손가락 센서 기술 개발
   경북대학교 - 교육연구및학생지도경비(연구영역)
   Role: 연구책임(PI) PI: Saekwang Nam (KNU)
   Management Agency: 경북대학교 Funding to Nam's Lab: 9.25M KRW
   Dates: Mar 1, 2024 – Feb 28, 2026 Note: 3책5공 제외사업

ACADEMIC SERVICES

2026	**Local Chair**, 2026 Asia Haptics Conference
2026	**Extended Abstracts Chair**, The 9th IEEE-RAS International Conference on Soft Robotics (RoboSoft 2026)
2026	**Member of the Organizing Committee**, The 4th Korean Haptics Society PI Symposium (제4회 한국햅틱스학회 PI Symposium 준비위원)
2025-26	**Director of International Cooperation**, Korean Haptics Society (한국햅틱스학회 국제협력 이사)
2025	**Member of the Organizing Committee**, The 3rd Korean Haptics Society PI Symposium (제3회 한국햅틱스학회 PI Symposium 준비위원)

TEACHING EXPERIENCE

2026	Spring	Data Computing at GSDS KNU
2026	Spring	Data-based Robotics at GSDS KNU
2026	Spring	Fundamentals of Robot & Mechanical Engineering at the department of Robotics KNU
2025	Fall	Data Computing at GSDS KNU
2025	Fall	Pragmatic Robot Programming at GSDS KNU
2025	Spring	Data Programming at GSDS KNU
2025	Spring	Data-based Robotics at GSDS KNU
2024	Fall	Data Process and Modeling at GSDS KNU
2024	Fall	Data Commputing at GSDS KNU
2024	Spring	Data Science Computing I at GSDS KNU
2024	Spring	Data Science Convergence Research at GSDS KNU

TEACHING AND MENTORING

Jan. 2012 - Mar. 2013	He had taught **Japanese** to undergraduate students at UCSD for **four quarters**. Although he was not a native Japanese speaker, the lecturers kindly gave me continuous chances to have the TA role based on students' evaluations and the Japanese Language Proficiency Test certificate with the highest level.
November 2020	He taught the concept of impedance measurements, how to measure the impedance, and how to find the equivalent electrical circuit based on the electrochemical impedance spectroscopy to **Carolin Anna Both**, a student at Friedrich-Abel-Gymnasium, for one week.
May 2022	He introduced the sensing principle of the transparent moisture sensor and ways of impedance measurement and microscope usage to **Tsujii Manon**, a Gymnasium student, for a week.
Dec. 2022 - May 2023	**Naoki Shitanda** came to the University of Bristol as a visiting undergraduate student from Kyoto University. During his time there, Saekwang advised on sensor fabrication, the use of a robot arm and the Robot Operating System. Together, they participated in the 2023 RoboSoft manipulation competition.
May 2023 - Aug. 2023	He assisted **Toby Jack**, an MSc student in the Biorobotics program at the University of Bristol, in constructing a robotic system designed to measure and evaluate the softness of fruit using a soft tactile sensor.
Nov. 2023 - present	He is officially supervising **Chenghua Lu**'s doctoral research at the University of Bristol.

SKILLS

Software	Robot Operating System, MoveIt!, Gazebo, Matlab, Python, LabVIEW, SolidWorks, COMSOL Multiphysics, LaTeX, C/C++/C#, Unity, Origin, NX
Hardware	UR10, Robotiq gripper, Laser lithography, Metal sputter, Optical microscope, Scanning electron microscope, Laser scanning vibrometer, Phantom haptic devices, Data acquisition board, 3D optical profilometer, Signal generators, 3D printers, Laser cutters
Language	English (*fluent*), Korean (*native*), Japanese (*excellent*), German (*beginner(A2))*

TALKS AND MEDIA HIGHLIGHTS

Jan. 4th 2024	Invited talk in Korea University of Technology and Education
Jan. 3rd 2024	Invited talk in Pusan National University
Dec. 21st 2023	Invited talk in Sookmyung Women's University
Dec. 19th 2023	Invited talk in Electronics and Telecommunications Research Institute
Dec. 18th 2023	Invited talk in Gwangju Institute of Science and Technology
Oct. 18th 2022	Invited talk titled "Current in Robot Fingers" to the Mechanical Engineering Department in Republic of Korea Naval Academy
Aug. 3rd 2022	Invited talk titled "Optimizing a Viscoelastic Finite Element Model to Represent the Dry, Natural, and Moist Human Finger Pressing on Glass" to Prof. Chun's lab. in Korea University Sejong Campus
Dec. 15th 2021	A report titled "The artificial intelligence in the sense of touch: The present and future of haptic technologies" has been published in KIC Europe Issue 2021/12: Artificial intelligence & Robotics (Korean).
May 27th 2021	A method to optimize the parameters in a model using measured data and a case example in modeling finger-pressing behaviors, organized by Korean Scientist and Engineers Association in Germany (Online presentation)
May 24th 2021	Designing, Sensing, and Data-processing for Understanding Human Fingers, hosted by the Embodied Dexterity Group at University of California Berkeley
November 2020	He was invited for the interview with Tech GangJeong, a youtube channel focusing on new technologies. He introduced the procedures of doctoral studies in Max Planck Institute, my research, and current challenges in object grasping by robots.
Oct. 17th 2020	Reasons for the secure object-grasping by fingers, organized by Korean Scientist and Engineers Association in Germany (Online presentation)
October 2020	Max Planck Institute for Intelligent Systems published an article on how well several members in our department participated in the EuroHaptics conference in the COVID-19 outbreak. News of receiving the Best Poster Prize featured in the article.
May 2017	HelloDD published an article about the importance of the Technical Research Personnel Program, an alternative means of conducting mandatory military service in Korea.
January 2017	Chungcheong Today published Saekwang's opinion about the importance of developing path-breaking technologies.
June 2014	Many broadcasting companies in South Korea released news about the development of transparent tactile sensor array (KBS,Daejeon MBC, TJB, Yonhap News, YTN, and Arirang).

REFERENCES

Name	Position and affiliation	E-mail
Nathan F. Lepora	Professor at University of Bristol and Bristol Robotics Laboratory	n.lepora@bristol.ac.uk
Katherine J. Kuchenbecker	Director of Haptic Intelligence Department at Max Planck Institute for Intelligent Systems	kjk@is.mpg.de
Ki-Uk Kyung	Professor in Mechanical Engineering at KAIST	kyungku@kaist.ac.kr